Unwrapping the Gift: Benefits of Computer Technology

Lecture 1

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The Ubiquity of Computers and the Rapid Pace of Change

Have you used a computer today?

You have if you used…

- an ATM,
- a cell phone,
- music player (iPod)
- a modern appliance, or
- an electronic device.

Negative impacts of new technology

Unemployment

Poor customer service

Crime

Alienation

Loss of Privacy

Some Positive Impacts of New

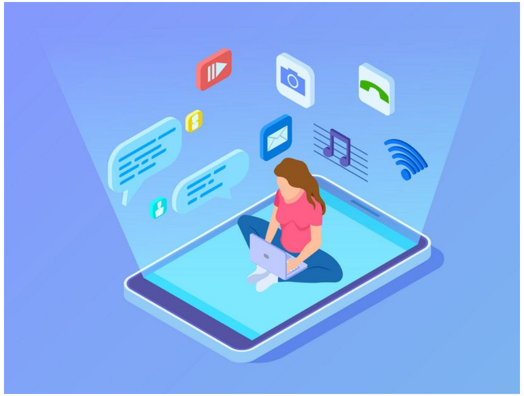
Convenience/ less effort

New types of jobs.

More options for transactions

Some Positive Impacts of New

Improvements in crime-fighting.



New forms of entertainment

New ways to communicate

Appreciating the Benefits

1. Benefits computers bring to **communication.**

2. Benefits computers bring to **transportation,** especially within our communities.

3. Benefits computers bring to **education** and training as they aid local schools.

4. Benefits of computers in **fighting crime**.

5. Benefits computers bring to **medicine**.

6. Benefits computers bring to the **disabled** in schools and the workplace.

Benefits to communication

- ❑ Non-invasive; read at recipients' convenience .
- ❑ Time-saving
- ❑ Access to immense (often timely) amounts of information
- ❑ Information on WWW can be in different formats such as text, graphics, video or sound.
- ❑ Much information on the web is "free" and can be accessed by all.
- ❑ Public forums not limited to geographic boundaries
    - Traditional boundaries are no obstacle
- ❑ More independence - people can do more themselves –
- ❑ Remote access available

Benefits to Transportation

- Navigation : Prevent from getting lost
- Diagnostics - e.g. chips that collect information about engine behaviour
- Safety : Anti-lock Braking System and Truck roll-over prevention
- Hybrid engines-New technologies, such as a warning system that alerts the driver he/she is veering off the road and at the same time corrects the drivers steering.
- Fuel efficiency
- Traffic pattern studies

Benefits to Education

- ❑ Language acquisition - Teach people to speak a foreign language
- ❑ Spelling
- ❑ People who are illiterate can learn in the comfort of their own home, without feeling embarrassed.
- ❑ Distance learning- . Students can learn and be taught from their own home without actually going to a school. (ex. Rural children unable to attend school due to distance or money)
- ❑ Simulations : Help train people to do a specific job.
- ❑ Speech recognition & synthesis (e.g. for learning a foreign language)

Benefits in fighting crime

- Improved crime reporting (e.g. information about stolen art may be posted)
- Faster search of arrest records and fingerprint files
- Remote access to records and reports (e.g., from patrol cars)
- Fingerprint matching has become easier with the introduction of there computer networks and databases. This task, as well as others, have become less time consuming and less painstaking
- Access to numerous databases
- Improved monitoring and surveillance equipment

Thank You

Unwrapping the Gift: Benefits of Computer Technology

Lecture 2

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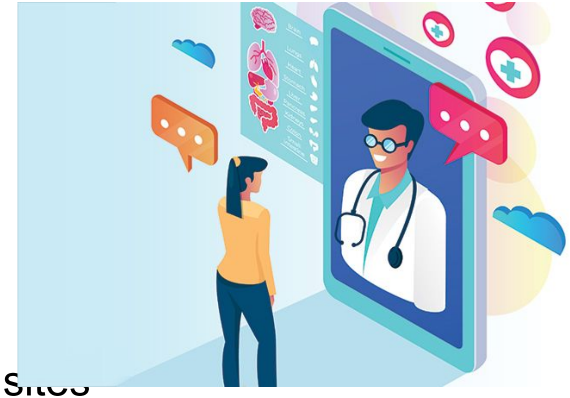
Benefits to medicine

- Sophisticated medical imaging
- Reduced surgery time
- Training surgeons
- Faster recovery time (e.g. minimally invasive surgery)
- Patient monitoring
- Improved treatments
- Patient records
    - More legible
    - Better organized
    - Accessible by many sites, simultaneously
    - Available immediately

Benefits to medicine

- ❑ Diagnosis
  - ○ Streamlined diagnosis
  - ○ Improved accuracy
  - ○ Reduced need for biopsy
  - ○ Improved screening and predictions
- ❑ Telemedicine
  - ○ Broad access to medical help
  - ○ Less expensive
  - ○ Expands consultations beyond rural/remote sites



Benefits to disabled

- Disability-specific computer applications (e.g., a braille printer, large size print).
- Control of household and workplace appliances via speech (people who can't use their hands.)
- Mobility
- Control of artificial limbs
- Improved vision (e.g. large font sizes)
- Access to adaptive educational equipment
- Improved communication (e.g. speech input for people who can't write, speech synthesis for those who can't talk)
- Opportunity to return to work

Benefits – Automation

- Organize and access large inventories
- Cost-saving
- Time-saving
- Improved customer satisfaction
- Tedious, monotonous work done by machines
- Improved workplace safety

Identifying people and products

- E.g. Bar codes & smart cards –
- Time-saving
- Accuracy
- Ease-of-use
- Cost-saving
- More secure
- Customizable
- Reliable

Monitoring Environment and Materials

- ❑ E.g. Bar codes & smart cards –
- ❑ Time-saving
- ❑ Accuracy
- ❑ Ease-of-use
- ❑ Cost-saving
- ❑ More secure
- ❑ Customizable
- ❑ Reliable

Benefits of Computers in Reducing Paper Use and Garbage

- ❑ Send/receive digital documents instead of hardcopy.
- ❑ Read, write, and edit online.
- ❑ Cost-saving; smaller storage needed.
- ❑ Creation of toxic wastes at paper mills reduced.

Other Benefits

- ❑ Tracking
- ❑ Rare plants and animals
- ❑ Migration habits
- ❑ Stolen goods
- ❑ Lost pets or people

Thank You

A GIFT OF FIRE THIRD EDITION - SARA BAASE
CHAPTER 2: PRIVACY

Lecture 1
Privacy and Computer Technology

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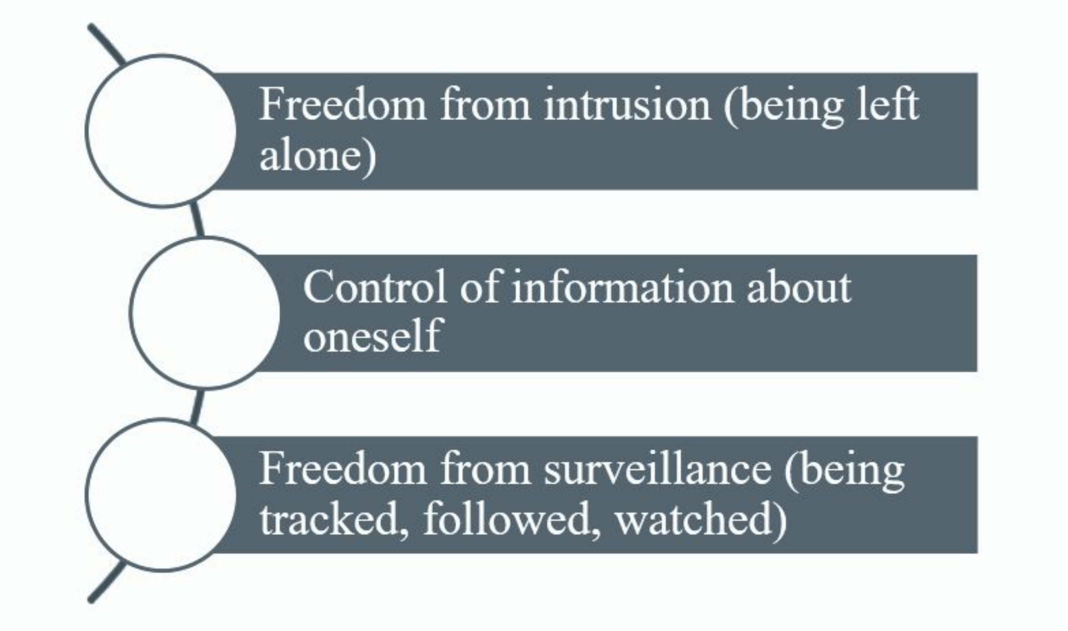
PRIVACY AND COMPUTER TECHNOLOGY

- Three Key Aspects of Privacy:



PRIVACY RISKS AND PRINCIPLES

Privacy threats come in several categories:

- ❏ Intentional, institutional uses of personal information
- ❏ Unauthorized use or release by “insiders”
- ❏ Theft of information
- ❏ Inadvertent leakage of information
- ❏ Our own actions

PRIVACY AND COMPUTER TECHNOLOGY

New Technology, New Risks:

- Government and private databases
  - Thousands of databases containing personal information about us.
  - Profiles of our personal information could be created easily.
- Sophisticated tools for surveillance and data analysis
  - Cameras, GPS, cell phones
- Vulnerability of data
  - Leaks of data happens, existence of data presents a risk.

PRIVACY RISKS AND PRINCIPLES

New Technology, New Risks – Examples:

- Search query data
  - Search engines collect many terabytes of data daily.
  - Data is analyzed to target advertising and develop new services.
  - Who gets to see this data? Why should we care?
- Smartphones
- Location apps
- Data sometimes stored and sent without user’s knowledge

PRIVACY RISKS AND PRINCIPLES

New Technology, New Risks – Summary of Risks:

- Anything we do in cyberspace is recorded.
- Huge amounts of data are stored.
- People are not aware of collection of data. Software is complex.
- Leaks happen.
- A collection of small items can provide a detailed picture.
- Re-identification has become much easier due to the quantity of information and power of data search and analysis tools.

PRIVACY RISKS AND PRINCIPLES

New Technology, New Risks – Summary of Risks:

- ❑ If information is on a public Web site, it is available to everyone.
- ❑ Information on the Internet seems to last forever.
- ❑ Data collected for one purpose will find other uses.
- ❑ Government can request sensitive personal data held by businesses or organizations.
- ❑ We cannot directly protect information about ourselves. We depend upon businesses and organizations to protect it.

PRIVACY AND COMPUTER TECHNOLOGY

Terminology and principles for data collection and use:

- ❑ Invisible information gathering
  - ❑ Collection of personal information about someone without the person's knowledge
  - ❑ Unauthorized software, Cookies, ISP providers •
- ❑ Secondary use
  - ❑ Use of personal information for a purpose other than the one it was provided for .
  - ❑ Data mining
    - ❑ Searching and analyzing masses of data to find patterns and develop new information or knowledge
  - ❑ Computer matching
    - ❑ Combining and comparing information from different databases (using social security number, for example, to match records)

Computer profiling

- Using data in computer files to predict likely behaviors of people
- Analyzing data in computer files to determine characteristics of people most likely to engage in certain behavior
- Businesses find new consumers
- Government detects fraud and crime activities
- Businesses engage in profiling to determine consumer propensity toward a product or service.
- Government agencies use profiling to create descriptions of possible terrorist

Principles for Data Collection and Use:

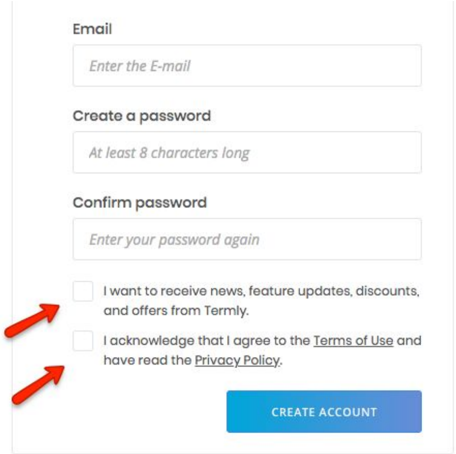
- ❑ Informed consent
- ❑ Opt-in and opt-out policies
- ❑ Fair Information Principles (or Practices)
- ❑ Data retention - Data retention is the storing of information for a specified period.

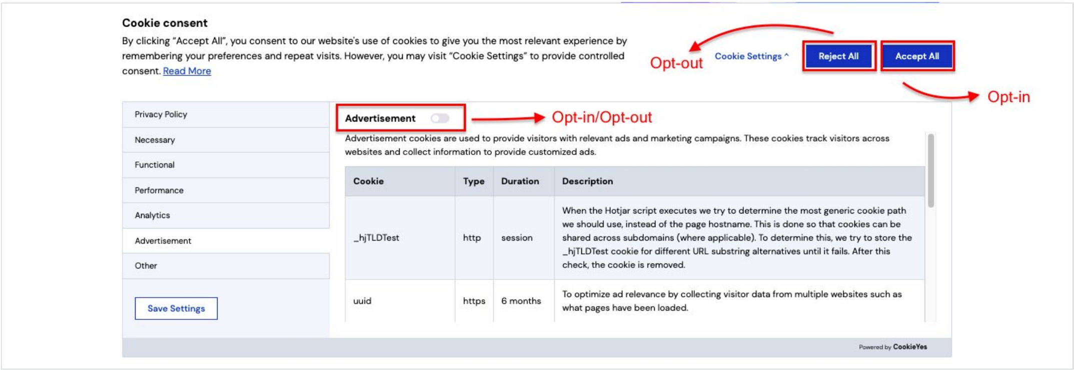
Principles for Data Collection and Use:

- Informed consent :
  - People should be informed about the data collection and use policies of a business or organization,
  - They can then decide whether or not to interact with that business or organization.
- Opt-in and opt-out policies :
  - Opt-in :
    - Opting in means that a user will take an affirmative action to offer their consent.
    - The most common way businesses implement opt-in methods is through checkboxes. When presented with a checkbox, the user must take action to check the box, which denotes their consent.

Principles for Data Collection and Use:

- Opt-in policies:
  - Opting in can be used in a variety of situations, including subscribing to email and newsletter mailing lists, accepting cookie use, and agreeing to legal policies.





Principles for Data Collection and Use:

- Opt-out policies :
  - Opting out means a user takes action to withdraw their consent.
  - Not choosing to subscribe to newsletters, unticking a previously ticked checkbox, not consenting to save personal details, rejecting the use of cookies, etc. are some examples of opt-out.

Principles for Data Collection and Use:

- Fair Information Principles (or Practices)
  - Inform people when you collect information about them, what you collect, and how you use it.
  - Collect only data needed
  - Offer a way for people to opt out from mailing lists, advertising, and other secondary uses.
  - Offer a way for people to opt out from features and services that expose personal information.
  - Keep data only as long as needed
  - Protect security of data (from theft and from accidental leaks). Provide stronger protection for sensitive data.
  - Policies for responding to law enforcement

Thank You

A GIFT OF FIRE THIRD EDITION - SARA BAASE
CHAPTER 2: PRIVACY

Lecture 2
"Big Brother is Watching You”

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"BIG BROTHER IS WATCHING YOU"

- Databases:
  - Government agencies
    - Collect many types of information
    - Ask business to report about consumers
    - Buy personal information from sellers
  - Main publicized reason: data mining and computer matching to fight terrorism
  - Private information can be used to:
    - Arrest people
    - Jail people
    - Seize assets

"BIG BROTHER IS WATCHING YOU"

- Databases:
  - Millions of crime suspects are searched in government databases
  - Shift from presumption of innocence to presumption of guilt
  - Computer software characterizes suspects
    - Innocent people are sometimes subject to embarrassing searches and expensive investigations and to arrest and jail.

"BIG BROTHER IS WATCHING YOU"

Small Sampling of Government Databases with Personal Information

- What data does the government have about you?
  - Tax records
  - Medical records
  - Marriage and divorce records
  - Property ownership
  - Welfare records
  - School records
  - Motor vehicle records
  - Voter registration records
  - Books checked out of public libraries
  - People with permits to carry firearms
  - Applications for government grant and loan programs
  - Professional and trade licenses
  - Bankruptcy records
  - Arrest records

"BIG BROTHER IS WATCHING YOU"

- Some constitution articles (laws) protect privacy.
- Modern surveillance techniques are redefining expectation of privacy.
- In some countries:
  - No court order or court oversight needed to get one’s private information.
  - 2003-2005 report found "widespread and serious misuse" of the FBIs (**Federal Bureau of Investigation**)national security letter authorities.

"BIG BROTHER IS WATCHING YOU"

Two key problems arise from new technologies:

- ❑ Much of our personal information is no longer safe in our homes; it resides in huge databases outside our control.
- ❑ New technologies allow the government to search our homes without entering them and search our persons from a distance without our knowledge

"BIG BROTHER IS WATCHING YOU"

Video Surveillance:

- ❑ Security cameras in Shopping centers, malls, banks, etc.
- ❑ Cameras alone raise some privacy issues.
- ❑ When being combined with face recognition systems, they raise more privacy issues.
    - ➢ Increased security
    - ➢ Decreased privacy

Thank You

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CHAPTER 2: PRIVACY

Lecture 3
"DIVERSE PRIVACY TOPICS ”

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Marketing, Personalization and Consumer

- Targeted and personalized marketing (business, political parties, etc)
- Paying for consumer information Examples -
    - Targeting ads to users by scanning their emails! Like Gmail.
- Data firms and consumer profiles
    - Companies (firms) that collect information about individuals
    - These firms sell data to other companies for marketing purposes.
- Data mining is being used
- Credit records might be sold to different parties

Social Networks

- What we do
  - Post opinions, gossip, pictures, “away from home” status
- What they do
  - New services with unexpected privacy settings

Location Tracking:

- Global Positioning Systems (GPS) -
  - computer or communication services that know exactly where a person is at a particular time
- Cell phones and other devices are used for location tracking

Stolen and Lost Data:

- Hackers
- Physical theft (laptops, thumb-drives, etc.)
- Requesting information under false pretenses
- Bribery of employees who have access

What We Do Ourselves

- Some people do not know or understand enough how the web works in order to make good decisions about what to put there.
- Some people do not think carefully.
- People often want a lot of information about others but do not want others to have access to the same kind of information about themselves.
- Our cell phone and email messages reside on computers outside our home or office.
- We have no direct control over such files.

What We Do Ourselves

- There have been many incidents of exposure of emails for politicians, businessmen, etc
- Personal information in blogs and online profiles
- Pictures of ourselves and our families
- File sharing and storing
- Is privacy old-fashioned?
  - Young people put less value on privacy than previous generations.
  - May not understand the risks.

Public Records: Access vs. Privacy:

- ➢ Public Records -
  - ○ Governments maintain "public records,"
  - ○ Records available to general public (bankruptcy, property, and arrest records, salaries of government employees, etc.)
- ➢ Identity theft can arise when public records are accesses.
- ➢ How should we control access to sensitive public records?

Children (privacy and safety)

- The Internet
  - Not able to make decisions on when to provide information
  - Vulnerable to online predators
- Parental monitoring
  - Software to monitor Web usage
  - Web cams to monitor children while parents are at work
  - GPS tracking via cell phones or RFID

Children (privacy and safety)

- At what age does web monitoring become an invasion of the child’s privacy?
- Should parents tell children about the tracking devices and services they are using?
- Informed consent is a basic principle for adults.
- At what age does it apply to children? Will intense tracking and monitoring slow the development of a child’s responsible independence? Will parents rely more on gadgets than on talking to their children?

National ID System:

- Social Security Numbers
  - Too widely used
  - Easy to falsify
- A new national ID system - Pros
  - would require the card
  - harder to forge
  - have to carry only one card
- A new national ID system - Cons
  - Threat to freedom and privacy
  - Increased potential for abuse

Thank You

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CHAPTER 2: PRIVACY

Lecture 4
"PROTECTING PRIVACY”

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Technology and Markets:

Awareness:

Most people have figured out now you can't do anything on the web without leaving a record.

- We can decide to what extent we wish to use privacy-protecting tools.
- We can be more careful about the information we give out, and the privacy policies of sites we use or visit.

Technology and Markets:

Privacy-enhancing technologies for consumers

- New applications of technology often can solve problems that arise as side effects of technology.
- Example: cookie disablers, blocking pop-up ads, scanning PCs for spyware, etc.
- Using usernames and passwords for Blogs visitors (family , friends, etc.)

Technology and Markets:

Encryption

- Information sent to and from websites can be intercepted.
- Someone who steals a computer or hacks into it can view files on it .
- Encryption is a technology that transforms data into a form that that is meaningless to anyone who might intercept or view it.
- Encryption generally includes a coding scheme, or cryptography algorithm, and specific sequences of characters (digits or letters).

Cryptography is the art and science of hiding data in plain sight.
—Larry Loen

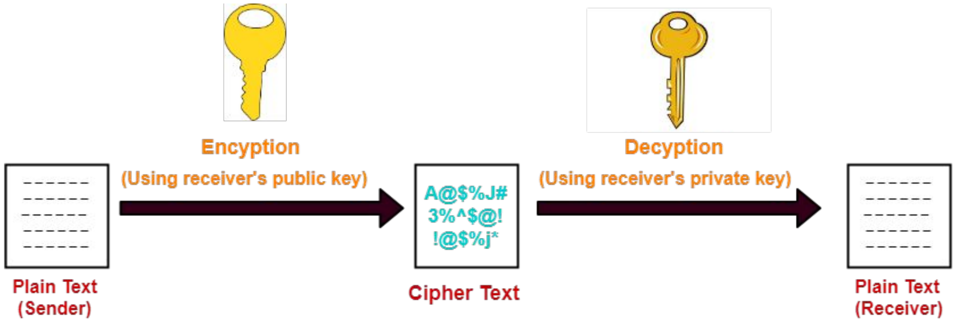
Technology and Markets:

Encryption

- Public-key cryptography: An encryption scheme, where two related keys are used.

    I. A key to encrypt messages.

    II. A key to decrypt messages.



Asymmetric Key Cryptography

Technology and Markets:

Business tools and policies for protecting privacy

- A well-designed database for sensitive information includes several features to protect against leaks, intruders, and unauthorized employee access.
  - Each person with authorized access to the system should have a unique identifier and a password.
  - A system can restrict users from performing certain operations, such as writing or deleting, on some files. User IDs can be coded so that they give access to only specific parts of a record.
  - For example, a billing clerk in a hospital does not need access to the results of a patient’s lab tests.

Technology and Markets:

Business tools and policies for protecting privacy

- The computer system keeps track of information about each access, including the ID of the person looking at a record and the particular information viewed or modified. This is an audit trail that can later help trace unauthorized activity.
- Website operators pay thousands, sometimes millions, of dollars to companies that do privacy audits. Privacy auditors check for leaks of information, review the company's privacy policy and its compliance with that policy, evaluate warnings and explanations on its website that alert visitors when the site requests sensitive data, and so forth.
- Some large companies like IBM and Microsoft, use their economic influence to improve consumer privacy on the web, by removing ads from their web sites.

Rights and laws:

Warren and Brandeis: The inviolate personality -

- Warren and Brandeis criticized newspapers especially for the gossip columns.
- People have the right to prohibit publications of facts (and photos) about themselves.
- The kinds of information of most concern to them are personal appearance, statements, acts, and interpersonal relationships (marital, family, and others)
- Warren and Brandeis say privacy is distinct and needs its own protection.

Rights and laws:

Thomson: Is there a right to privacy? -

- Thomson argues the opposite point of view.
- There is no violation of privacy without violation of some other right, such as the right to control our property or our person, the right to be free from violent attacks, the right to form contracts(and expect them to be enforced).

Rights and laws:

Applying the theories:

- Many court decisions since Warren and Brandeis article, have taken their point of view.
- A person may win a case if someone published his/her consumer profile.
- Warren and Brandeis (and court decisions) allow disclosure of personal information to people who have an interest in it.
- An important aspect: consent

Rights and laws:

Transactions

- Privacy includes control of information about oneself.
- How to apply privacy notions to transactions, Which involve more than one person?

Rights and laws:

Ownership of personal data

- People should be given property rights in information about themselves.
- But some activities and transactions involve at least two people, each of whom would have claims to own the information about the activity.
- Can we own our profiles (collection of data describing our activities, purchase, interests, etc.) ?
- We cannot own the fact that our eyes have a certain color

Rights and laws:

Regulation

- Technical tools for privacy protection, market mechanisms, and business policies are not perfect.
- Regulation is not perfect either.
- Some Regulations may be so expensive and difficult to apply.
- Example: Health Insurance Portability and Accountability Act (HIPAA)- protect sensitive patient health information from being disclosed without the patient’s consent or knowledge.

Rights and laws: Contrasting Viewpoints:

Free Market View

- Freedom of consumers to make voluntary agreements
- Diversity of individual tastes and values
- Response of the market to consumer preferences
- Usefulness of contracts

Rights and laws: Contrasting Viewpoints:

Consumer Protection View

- Advocates of strong privacy regulation emphasize the unsettling uses of personal information
- Costly and disruptive results of errors in databases
- Ease with which personal information leaks out, via loss, theft, and carelessness
- Consumers need protection from their own lack of knowledge, judgment, or interest

Communications

Wiretapping and Email Protection:

Telephone

- 1934 Communications Act prohibited interception of messages
- 1968 Omnibus Crime Control and Safe Streets Act allowed wiretapping and electronic surveillance by law-enforcement (with court order)

Communications

Wiretapping and Email Protection:

- Electronic Communications Privacy Act of 1986 (ECPA) extended the 1968 wiretapping laws to include electronic communications, restricts government access to e-mail

Thank You

A GIFT OF FIRE THIRD EDITION - SARA BAASE
CHAPTER 3: Technology & Free Speech

Lecture 1

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Changing paradigms

- Regulatory paradigms
  - Communications technologies differ with respect to constitutional protections
  - Also with respect to government regulation
- The traditional three-part framework
  - Print media (Newspapers, magazines)
  - Broadcast media (Television, radio
  - Common carriers (Telephone, telegraph, and the postal system)

Changing paradigms

- Print media (Newspapers, magazines)
  - Has the strongest protections for free speech
- Broadcast media (Television, radio)
  - Has protections, but govt controls structure of industry + content
  - E.g. Canadian content regulations, restriction of some words
- Common carriers (Telephone, ISPs?)
  - Content not controlled and carrier not responsible for content

Changing paradigms

- Consider new technologies of the past 20 years
  - Bulletin-board systems (BBS)
  - World Wide Web
- How do these fit into the “print-broadcast-carrier” framework?
  - Not exactly print media
  - Not exactly broadcast media
  - Not exactly common carrier

Offensive Speech

Examples of such speech can include

- ❑ Political or religious speech
- ❑ Pornography
- ❑ Sexual or racial slurs
- ❑ Nazi (or White Supremacist) materials
- ❑ Libellous (false and damaging) statements
- ❑ Abortion information
- ❑ Alcohol advertisement

Censorship

- Censorship is the suppression of speech, public communication, or other information.
- This may be done on the basis that such material is considered objectionable, harmful, sensitive, or "inconvenient".
- Censorship can be conducted by governments, private institutions and other controlling bodies.
- Governments and private organizations may engage in censorship. Other groups or institutions may propose and petition for censorship.
- General censorship occurs in a variety of different media, including speech, books, music, films, and other arts, the press, radio, television, and the Internet for a variety of claimed reasons including national security, to control obscenity, pornography, and hate speech, to protect children or other vulnerable groups, to promote or restrict political or religious views, and to prevent slander and libel.

Offensive speech & censorship

- One approach used in the US
  - Limit Internet access in libraries and schools.
  - Accomplish this via “filtering software”
  - “Funding” control (any school or library receiving US federal funds must install such software on internet terminals)
- Example: Canadian company • Userful.com (Calgary)
- software does much, much more than just web filtering
- Is CIPA compliant
- Filters block sites containing child pornography, obscene material, anything deemed “harmful to minors”

Hate Speech

- Racist and hateful comments are offensive…
  - but not necessarily illegal.
- “hate speech” loosely refer to offensive discourse targeting a group or an individual based on inherent characteristics - such as
  - race,
  - religion or
  - gender - and that may threaten social peace.
- A judge has the authority to order removal of hate propaganda from a computer system available to public.

Hate Speech

- Canadian Human Rights Act
- Section 13: Applies to e-mail, web sites, and any other telecommunications activity.
- Prohibits messages likely to expose a person to hatred or contempt on any one of several categories.
  - Race
  - national or ethnic origin
  - Colour
  - Religion
  - Age
  - sexual orientation
  - marital status
  - family status
  - Disability
  - conviction for which a pardon has been granted

Challenges to Regulation

- For certain kinds of activity, a licence is required:
  - To practice law
  - To practice medicine (e.g., prescribe medication)
  - To practice engineering
  - To offer financial advice (e.g., investing in equities, futures, commodities)
- For other kinds of activity, no licence is required
  - To create downloadable, self-help legal software.
  - To publish newsletter / website about, or develop software for commodities and futures investing.

Censorship & the global net

- Internet technology has a global impact on free speech
- Can be used to avoid censorship
  - Global nature of Internet allows restrictions in one country to be circumvented by using networks in other, less restrictive countries. Such sites are usually posted in the US.
- Can be used to establish censorship
  - Global nature of Internet makes it easier for one nation to impose restrictive standards on others.
- eBay bans some items for sale (i.e., hate-group memorabilia) because of different cultural standards in different countries.

Thank You

A GIFT OF FIRE THIRD EDITION - SARA BAASE
CHAPTER 3: Technology & Free Speech

Lecture 2

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Anonymity

- True anonymity
  - “Unlinkability”
  - Inability of anyone to determine the true author’s identity
  - Legally problematic

Anonymity

- Pseudonymity:
  - Pseudonymity is the near-anonymous state in which a user has a consistent identifier that is not their real name: a pseudonym.
  - public pseudonyms: link between pseudonym and human being is easy for the public to know or discover
  - non-public pseudonyms: link known by sysadmins, but no one else
  - unlinkable pseudonyms: link not known and not discoverable by service operators

Anonymous Re–mailers and Web Browsers

- Johan Helsingius set up the first well-known anonymous email service in Finland in 1993.
  - Users were not entirely anonymous
  - the system retained identifying information.
  - Return address stripped off and then forwarded to recipient
- To send anonymous email using a “re-mailer” service, one sends the message to the remailer, where the return address is stripped off and the message is re-sent to the intended recipient.
- Anonymous web browsers - hide user’s IP address, do not track or store activity

Anonymity vs. Community?

- Supporters of anonymity
  - Claim that it is necessary to protect privacy and free speech.
  - Helps those who are concerned about political or economic retribution.
    - Human rights workers
    - Victims of domestic violence

Anonymity vs. Community?

- Opponents of anonymity
  - Claim that it is anti-social and allows criminals to hide from law enforcement.
  - Threatens civil discourse (“marketplace of ideas”)
  - Examples:
    - Online harassment
    - Fraud

Spam

- Unsolicited, mass e-mail
  - Is cheap to senders but may impose costs on recipient’s time or the recipient’s online account (or both).
  - May contain objectionable content (political, commercial ads, solicitations for funds, pornography, etc.)
  - May contain a disguised return address.
  - May pass through filters.
- – Invades privacy.
- – Creates a financial and managerial burden on ISPs.

Spam: solutions (?!)

- Technology: Filters that screen out spam.
- Market pressure: Services that list spammers
- Small “microfee” for sending an email?
- requires consent for email marketing
- Vigilantism: Punish spammers

Thank You

A GIFT OF FIRE THIRD EDITION - SARA BAASE
CHAPTER 2: Computer Crime

Lecture 1

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Challenges wrt Computer Crime

- There are at least three:
- Prevention
- Detection
- Prosecution

Defining "Hacking": Phase one

Phase 1: Joy of Programming

- The early years (1960s and 1970s), when hacking was a positive term
- Hacker was a creative programmer who wrote clever code.
- First OSes and computer games were written by such hackers.
- Hackers were usually – but not always – high-school and college students.

Phase 2: From the 1970s to the

- Negative overtones appear in 1970s-90s
- Popular authors and media caused this use of term:
  - Described someone who used computers without authorization.
  - Sometimes to commit crimes.
- Early computer crimes were launched against business and government computers.
- Adult criminals began using computers
  - “White collar” crime
  - Organized crime increasingly responsible for computer break-ins

Phase 3: The growth of the Web and mobile devices

- Web era (mid 90s)
- Increased use of Internet by average people
- Attractive to criminals with basic computer skills.
- Crimes included the release of malicious code
- Unprotected computers are especially vulnerable
- Hacking for political motives increased

Phase 3: The growth of the Web and mobile devices

- Unsuspecting users may have their computers utilized to take part in a DDoS or fraud
- Minimal computer skills needed to create havoc
  - “Script kiddies”
    - Attackers who use tools / code written by others

Hacktivism

- Hacktivism is the use of hacking to promote a political cause
- Degree can range from mild to destructive
  - Defacing websites
  - Destroying data
  - Denial of service
- People who agree with the political or social position of the hackers will tend to see an act as “activism,” while those who disagree will tend to see it as ordinary crime (or worse).
- Some academic writers and political groups argue that hacktivism is ethical, that it is a modern form of civil disobedience.

Hacktivism

- ❑ Others argue that the political motive is irrelevant, or at the other extreme, that political hacking is a form of cyberterrorism.
- ❑ They think so:
  - ❑ Denies others their own freedom of speech.
  - ❑ Violates property rights.
  - ❑ Even some hacktivists reject web-site defacing as legitimate.activity.

Thank You

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CHAPTER 2: Computer Crime

Lecture 2

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Cyber Crimes

- ❑ Cybercrime is any criminal activity that involves a computer, networked device or a network.
- ❑ Cyber crime is a subset of computer crime.
- ❑ Definitions :
  - ○ Offences that are committed against individuals or group of individuals with a criminal motive
    - ■ to intentionally harm the reputation of the victim or
    - ■ cause physical or mental harm to the victim directly or indirectly, using modern telecommunication networks such a s internet.

TARGETS OF CYBER CRIMES

- Against individual property
  - Transmitting virus
  - Un-authorized control/access over computer
  - Intellectual property crimes
  - Internet time thefts
- Against Organization
  - Possession of un-authorized information
  - Cyber terrorism against the government organization
  - Distribution of pirated software,etc

TARGETS OF CYBER CRIMES

- Against Social at Large
  - Pornography (basically child pornography)
  - Trafficking
  - Financial crimes
  - Online gambling
  - Forgery
  - Sale of illegal articles

SOME COMMON CYBER CRIMES

- Computer Virus: A computer virus is a computer program or attaches itself to application programs or other executable system software causing damage to the files.
- Phishing: Phishing occurs when the perpetrator sends fictitious e-mails to individuals with links to fraudulent websites that appear official and thereby cause the victim to release personal information to the perpetrator.

SOME COMMON CYBER CRIMES

- Hacking: The act of penetrating or gaining unauthorized access to or use of data unavailable in a computer system or a computer network for the purpose of gaining knowledge, stealing or making unauthorized use of the data.
- Spoofing : Spoofing, as it pertains to cybersecurity, is when someone or something pretends to be something else in an attempt to gain our confidence, get access to our systems, steal data, steal money, or spread malware

SOME COMMON CYBER CRIMES

- Netsplonage: Netsplonage occurs when perpetrators back into online systems or individual PCs to obtain confidential information for the purpose of selling it to other parties.
- Cyber stalking: Cyber stalking refers to the use of the internet, email or other electronic communications device to stalk another person. It is an electronic harassment that involves harassing or threatening over a period of time.
- Cyber Terrorism: Cyberterrorism is the convergence of cyberspace and terrorism. It refers to unlawful attacks and threats of attacks against computers, networks and the information stored therein when done to intimidate or coerce a government or its people in furtherance of political or social objectives.

MOTIVES OF CYBER CRIMINALS

- ❑ Desire for entertainment
- ❑ Profit
- ❑ Infuriation or revenge
- ❑ Political agenda
- ❑ Sexual motivations
- ❑ Psychiatric illness

PREVENTION

- ❑ Use hard to guess passwords
- ❑ Use anti-virus software and firewalls-keep them up to date
- ❑ Don’t open email or attachments from unknown sources
- ❑ Backup your data

Thank You